

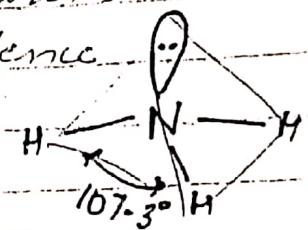
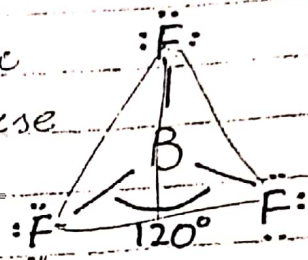
Inorganic Chemistry

Past papers SO (2013)

✓(1) (2nd Semester)

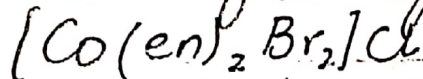
BF_3 molecule is planar but NH_3 is pyramidal.

BF_3 has planar shape while NH_3 has pyramidal because Boron has 3 electrons in the outermost shell while Nitrogen has 5 electrons in its outermost shell. Hence in NH_3 1 lone pair of electron is present and due to repulsion b/w lone pair & bond pair its shape become pyramidal.

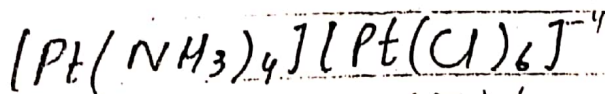


✓(2)

Name the following: (24)



Dibromobis(ethylenediamine) cobalt(III) chloride



tetraamineplatinum(II) hexachloroplatinate(IV)

✓(3) (257)

Give two limitations of VBT.

The VBT was fairly successful in explaining qualitatively the geometry of complexes, but still it has following limitations.

* It can't explain magnetic moment data of some complexes.

* Unable to explain nature of some of complexes of Copper.

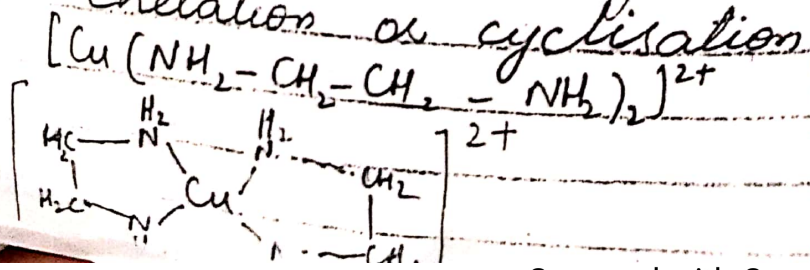
Draw M.O diagram of $Ni(CO)_4$

(4) (2413)

Chelates:

A chelating ligand is a bidentate or polydentate ligand which is attached to the same central metal atom by two or more of its donor atoms resulting in the formation of a complex having a strain-free ring structure. The complex having the ring structure is called chelate or chelated complex. Formation of chelate is called chelation or cyclisation.

Example:



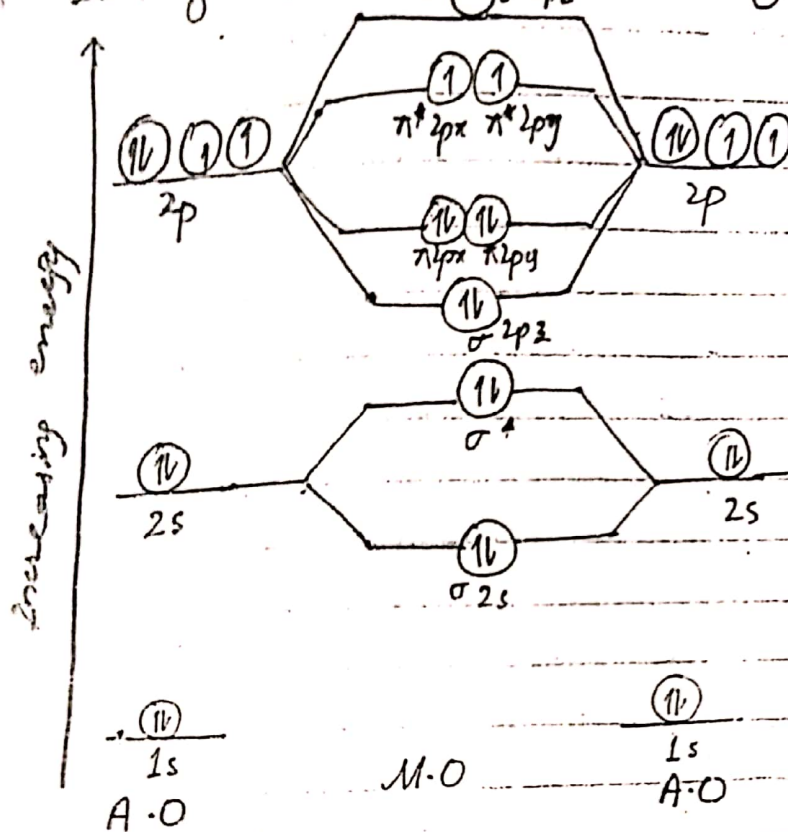
3

(6) (288)

Molecular orbital diagram of Oxygen:

The electronic configuration of Oxygen:

$1s^2, 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$



$$\text{Bond order} = \frac{1}{2} (8 - 4) = \frac{1}{2} (4) = 2$$

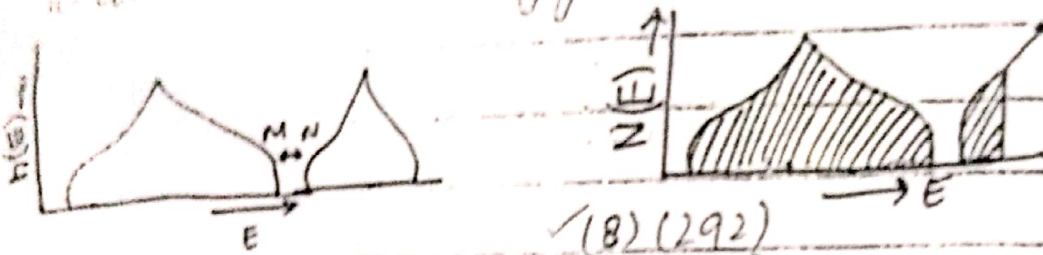
(7) (212)

Difference b/w $N(E)$ and $n(E)$ curves:

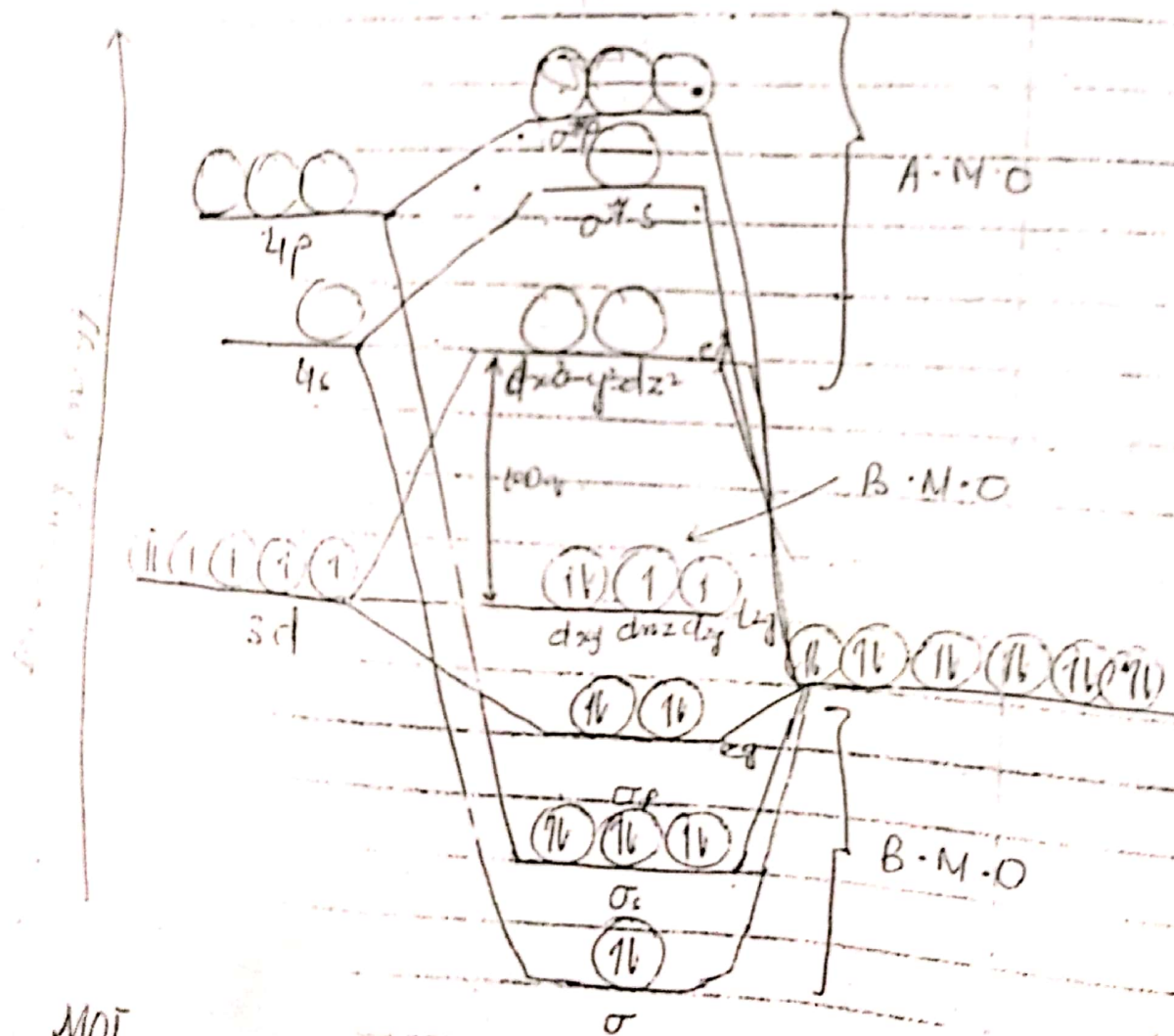
$n(E)$ curves are used to show the complete distribution of electrons b/w the range of energy level in an energy band. $N(E)$ curves showing the distribution of electron states over the energy range within a band.

$\int n(E) dE$ being the number of electrons per unit volume of material with energies b/w E and $(E + dE)$.

$N(E)dE$ gives the number of energy states with energy b/w E and $(E+dE)$



MOT diagram of $[\text{CoF}_6]^{3-}$ ion



MOT energy level diagram of $[\text{CoF}_6]^{3-}$ ion.

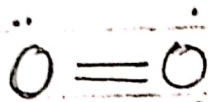
which produce a large Δ are called strong-field ligands and lie at the right end of the series.

(10)

O_2 is paramagnetic while O_2^{2-} is diamagnetic.

O_2 is paramagnetic because it has two unpaired electrons.

Lewis structure of O_2

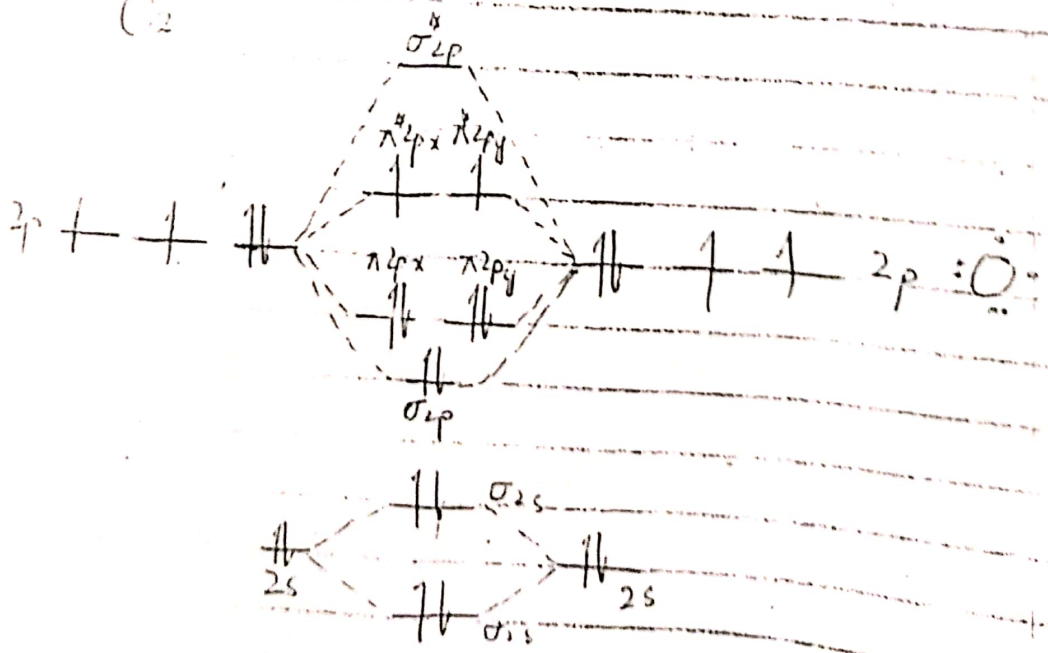


It shows that all the electrons in O_2 are paired, so oxygen should be diamagnetic.

Yet O_2 is paramagnetic.

According to MOT diagram of O_2 , first two electrons are unpaired. O_2 is paramagnetic.

O_2



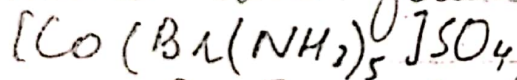
(2014)

(i)

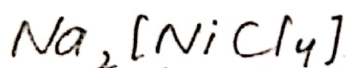
NOT diagram of $Mn_2(CO)_{10}$

✓(2)

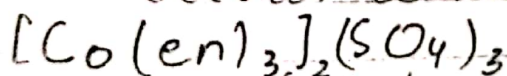
Name the following



Pentamminebromidocobalt(II) Sulphate



Sodium tetrachloronickelate(II)



Tris(ethylenediamine) cobalt(III)

Sulphate

✓(3) (255)

Outer transition or d-block elements shows variable oxidation states:

Outer transition element show variable oxidation states in their compounds.

Reason of variable oxidation

It is that there is a very small energy difference in between $(n-1)d$ and ns orbitals. As a result, $(n-1)d$ -orbitals as well as ns -orbitals take part in bond formation.

✓(4)(209)

Metallic bond on the basis of Band Theory:

The band theory of metallic bonding is based on the molecular orbital concept.

According to this, when two atomic orbitals combine, two molecular orbitals are formed. One has lower energy than the parent atomic orbitals, it is called bonding molecular orbital, the other has higher energy than the parent atomic orbitals, it is called anti-bonding molecular orbital.

✓(5)(213)

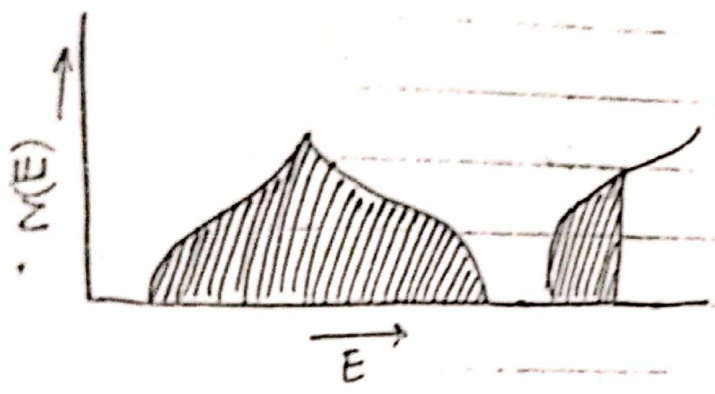
$N(E)$ Curves:

$N(E)$ curves showing the distribution of electron states over the energy range within a band.

$N(E)dE$ gives the no. of energy states with energy between E and $(E+dE)$.

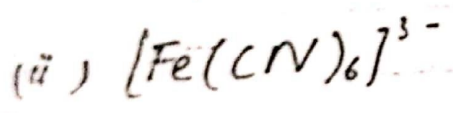
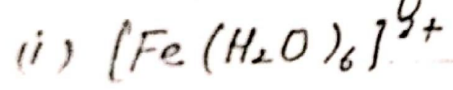
This may be separated by an energy gap. Alternatively, two $N(E)$

may overlap, and it is the shape of $N(E)$ curves which is of vital importance in conductivity consideration.



(6)

Explain CF splitting & E.C of d-orbital of central metal ion in following complexes:



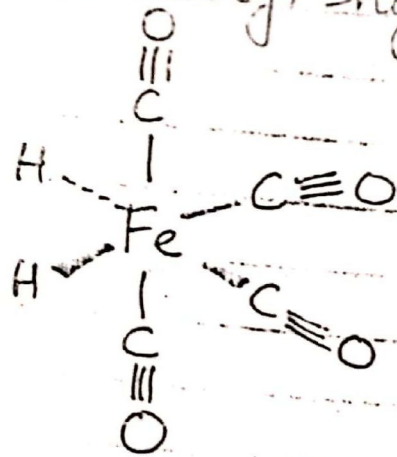
✓(17) (216)

N-Type semiconductors:

- It is an intrinsic semi-conductor which is obtained by doping the impurity, pentavalent impurity atoms such as antimony, phosphorus, arsenic etc. to pure germanium and silicon semiconductors.
- The impurity atoms added, provide extra electrons in the structure, and are called donor atoms.
- The electrons are majority charge carriers and holes are minority charge carriers.

✓(18)

Structure of $H_2Fe(CO)_4$
Iron tetracarbonyl hydride



✓(10)

O_2 is paramagnetic while O_2^{2-} is diamagnetic.

Repeat

Calculate
splitting
 d^4 low

(9)
The crystal field
magnitude for the
spin complexes:

(1)

CO is known as a π -acid-ligand:

It is a good π -acceptor due to empty π orbitals and a good sigma donor.

When bonding to a metal the ligand (CO) sigma donates to an empty d-orbital and the filled d-orbitals of the metal donate to the empty π orbitals of CO, back donation.

This occurs only when the metal has an oxidation state $\geq 3+$.

✓(2)

Difference b/w Semiconductor and Insulator:

Semiconductor

- Conducts the electric current less than conductor and greater than insulator.

- Has four valence electrons in its outermost orbit.

- Formed due to covalent bonding.

- Resistance is high.

e.g. Silicon, germanium

Insulator

- Does not conduct any current.

- Has eight valence electrons in its outermost orbit.

- Formed due to ionic bonding.

- Resistance is very high.

e.g. Mica, Paper

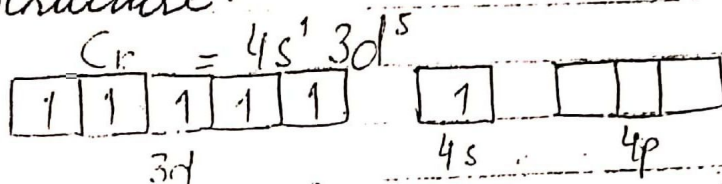
Structure of $[\text{Co}(\text{NH}_3)_6]^{3+}$ on the basis of MOT.

Repeat

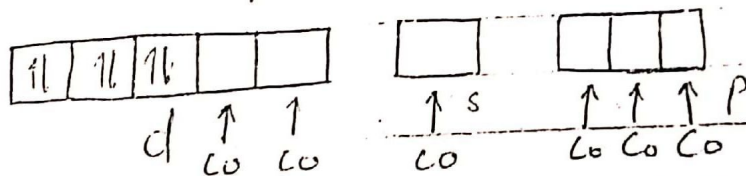
BF_3 is planar but NH_3 is pyramidal

Repeat

Show hybridization of $\text{Cr}(\text{CO})_6$ with structure:

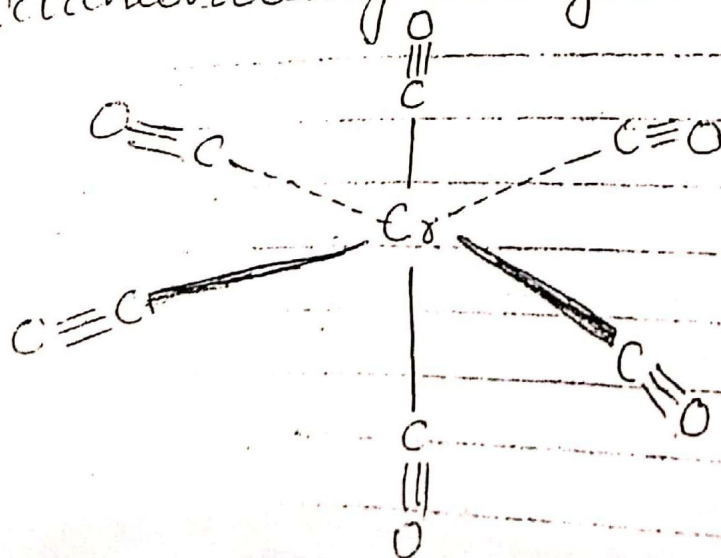


Since, CO is strong field ligand it will pair up the electrons



Hybridization = d^2sp^3

Diamagnetic in nature
Octahedral geometry

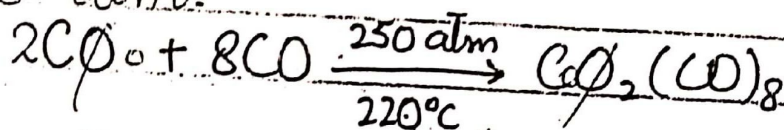


(6)

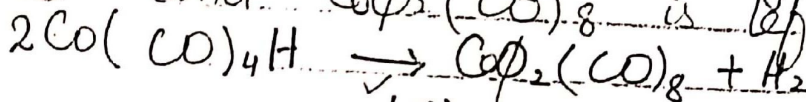
Two chemical methods for the preparation of $\text{Co}_2(\text{CO})_8$.

It can be prepared

- * By the reaction b/w CO and reduced metallic cobalt at 220° and 250 atm.



- * By the action of an acid on solution of $\text{Co}(\text{CO})_4\text{H}$. Hydrogen is evolved and $\text{Co}_2(\text{CO})_8$ is left behind.



(7)

MOT structure of $\text{H}_2\text{Fe}(\text{CO})_4$.

Repeat

(8)

Limitations of CFT.

The crystal field model is successful in explaining the formation, structure, colour and magnetic properties of coordination compounds to a large extent.

However this has some limitations.

→ This theory takes only d-orbitals of a central atom into account.

The s and p orbitals are not considered for the study.

→ The assumption that the interaction b/w metal-ligand is purely electrostatic cannot be said to be

very realistic.

→ This theory fails to explain the behaviour of central metal which cause large splitting while other show small splitting.
For example.

The theory has no explanation as to why H_2O is a stronger ligand as compared to OH^- .

(9)

O_2 is paramagnetic while O_2^{2-} is diamagnetic.

Repeat

(10)

Metallic bond on the basis of Band Theory:

Repeat

(2016)

(16)

✓(1)

Effect of impurities on conductivity of metals and semi-metals:

The regular array and field within a crystal of a metal can also be disturbed by small quantities of impurities. As a result conductivity may be greatly decreased by small amount of impurities.

For example:

The effect of adding manganese to copper:

% of Mn	0	1	2	3
Conductivity $/\Omega^{-1}\text{cm}^{-1}$	5.9×10^3	2.0×10^5	1.2×10^5	0.8×10^3

Conductivity of a semiconductor generally increases with the introduction of impurities because the impurities either provide extra electrons or have a deficiency of e^- which results in the introduction of holes. Both e^- and holes conduct.

✓(2)

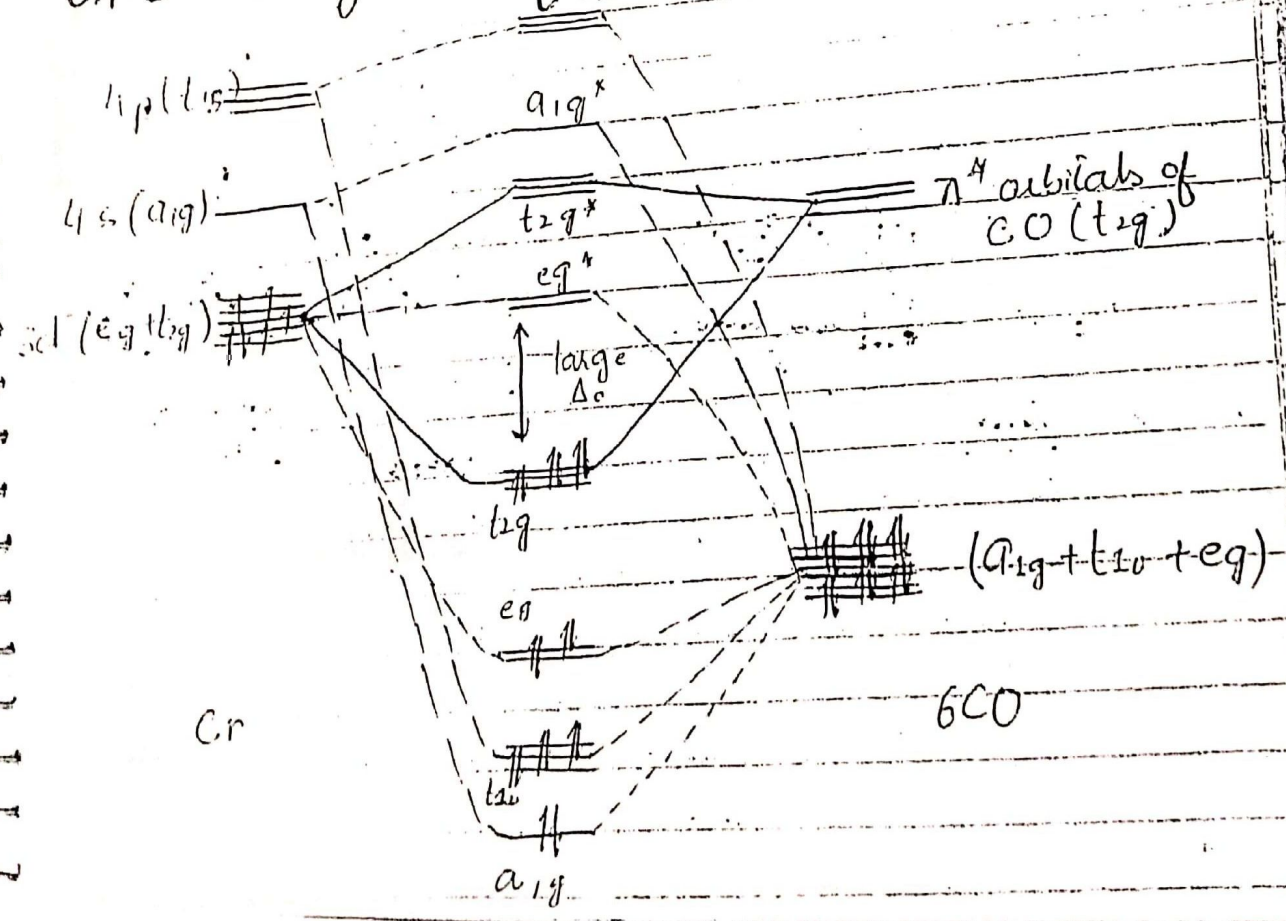
Best achievement of CFT:

(3)
Outer Transition element show variable oxidation states:

Repeat
(4)

Limitations of VBT:
Repeat

(5)
M.O diagram of $\text{Cr}(\text{CO})_6$

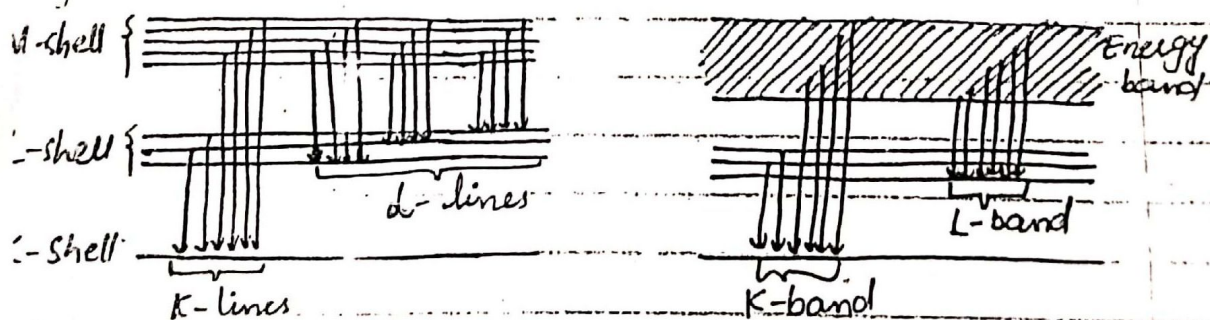


(5)

Soft X-ray spectra:

The existence of energy bands postulated in the band theory of metals is borne out by a study of soft X-ray spectra. It is indeed, a study of such spectra which provides experimental information about the bands.

When cooled metal is bombarded by beam of high-speed e^- , some e^- in the low energy level may be energized to be ejected and taking the vacant space (place) from higher level which cause emission of radiations in X-ray region. Electronic movement from K-shell give rise to K-lines in X-ray spectrum and e^- moving into L-shell gives L-lines and so on.



Soft X-ray spectra help to elucidate the arrangement of e^- in the energy band of metals and crystals.

(19)

(7)

Differentiate b/w $N(E)$ curves & $n(E)$ curves

Repeat

(8)

Differentiate b/w Semiconductors and insulators

Repeat

(9)

Define spectrochemical series

Repeat

(10)

O_2 is paramagnetic while O_2^{2-} is diamagnetic

Repeat

(2017)

(20)

Differentiate b/w (1)

Conductors	Semiconductors	Insulators
Easily conduct the electrical current	Conduct the electrical current less than conductors and greater than insulators	Does not conduct any current
Has only one valence electron in its outermost orbit	Has four valence electrons in its outermost orbit	Has eight valence electrons in its outermost orbit
Conductors formed using metallic bonding	Semiconductors are formed due to covalent bonding	Insulators are formed due to ionic bonding
Resistance is very small. It has positive temperature coefficient. Example Copper, aluminium etc	Resistance is high. It has negative temperature coefficients. Example Silicon, germanium etc	Resistance is very high. It has negative temperature coefficients. Example Mica, paper etc

(2)

CO is π -acceptor ligand.

Repeat

(3)

Limitations of VBT:
Repeat

✓(4)
M.O diagram of $[\text{Co}(\text{NH}_3)_6]^{3+}$
Repeat

✓(5)
Chelates:

Repeat

✓(6)
M.O diagram of O_2

Repeat

✓(7)

Difference b/w $N(E)$ and $n(E)$ curves

Repeat

✓(8)

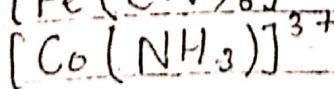
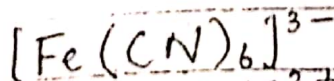
Inner orbital complexes:

→ Inner orbital complexes are coordinate compounds having a central metal atom that undergoes hybridization of atomic orbitals including the inner d-orbitals.

→ The d-orbitals involved in the hybridization are in lower energy level than s and p orbitals.

→ Composed of metal atoms that use inner shell d-orbitals for the hybridization in the central metal atom.

Example:



(9)

O_2 is paramagnetic while O_2^{2-} is diamagnetic

Repeat

(10)

Sodium is good conductor of electricity while silicon is a semi-conductor

why:

Sodium is a good conductor of electricity because its molecular structure consists of a lattice of positive ions encompassed by a "sea" of delocalised electrons, which can carry a charge and are mobile, hence, electric charges can flow through sodium easily through the electrons.

while silicon is a semiconductor because the four outer electrons in each atom are held strongly in covalent bonds hence there are few delocalised electrons and so, silicon is a poor conductor.